

Collections

Introduction to Programming

StayAhead Training; Sep, 2024

Introduction

Collections

- **Collection:**
 - A group of things in memory
 - Each member is accessible via the group
- A plural noun is indicative of a collection, e.g. clients, results, etc.

Lists

Collections

- **List:** the simplest and most common type of collection
- Lists are indexed; each element is accessed via its index
- A Python list can grow and shrink

List Creation

Collections

- Pseudocode, e.g.:
 - SET Names to ["Tom", "Dick", "Harry"]

List Creation

Collections

- `numbers = [2, 4, 6, 8]`
- `names = ["James", "George", "Olivia"]`

List with Various Types of Data Collections

- `things = ["James", 123, True]`

Getting Elements from a List

Collections

- Pseudocode, e.g.:
 - SET FirstName to Names[0]
 - OUTPUT FirstName to stdout
- Or, simply:
 - OUTPUT Names[0] to stdout

Getting Elements from a List

Collections

- `third_number = numbers[2]`

Setting Elements in a List (initial state)

Collections

Index	Element
0	2
1	4
2	6
3	8

Setting Elements in a List

Collections

- Pseudocode, e.g.:
 - SET Names[0] to "Thomas"

Setting Elements in a List

Collections

- `numbers[2] = 7`

Setting Elements in a List (after setting)

Collections

Index	Element
0	2
1	4
2	7
3	8